

LOCAL vs \$20/MO

The math nobody runs before buying a GPU.

8 slides · cost, speed, context, and where local actually wins

YOUR \$20 PLAN IS SUBSIDISED.

SemiAnalysis estimate: a hard-used \$20 ChatGPT Plus plan extracts about \$700 of API-equivalent compute. A maxed \$200 Pro plan: up to \$14,000.

One tracked user burned 10B tokens in 8 months: \$15,000 at API rates. About \$800 on subscriptions.

You are not paying for the compute. They are.

LOCAL LOSES BY 5x

Effective \$ per million tokens, 12-month total:

open-weight API	\$1.97
OpenAI API	\$6.90
Anthropic API	\$9.86
local, consumer	\$35.37

A casual user never pays the card back.

BREAK-EVEN IS ~36 MONTHS

12-month totals: OpenAI \$12,600 | local \$18,387

36-month totals: OpenAI \$37,800 | local \$32,870

Hardware amortisation is the whole story -

electricity is the small line.

Year one goes to the API. Year three goes to you.

ALWAYS-ON AGENTS

Past ~500M tokens/month the math flips:

- local \$1.04 - \$2.67 per M tokens
- API \$2,000+ per month
- payback 6 to 12 months

Also wins on: privacy, no rate caps, offline,
and zero vendor lock-in.

Steady, high volume - or nothing at all.

THE NUMBERS

first token: local 50-200ms | cloud 200-800ms

M5 Max 128GB: ~65 tok/s on 120B-class MoE

RTX 5060 Ti 16GB: 20-40 tok/s on 24B

context: Claude Pro 200K | Llama 3.3 128K

Plus caps: 160 messages per 3 hours

Local wins latency. Cloud wins the ceiling.

CLOSE IS NOT EQUAL

MMLU: Llama 3.3 70B 80% vs GPT-5.2 87%

Code: Qwen3-Coder 32B 92.7% HumanEval - parity

Agentic tasks over hours: local sub-30B collapses
to 31-38% where the frontier holds ~90%.

Error compounds per step, so long loops punish
the smaller model.

Great at bounded work. Weak at long horizons.

PICK BY VOLUME, NOT BY PRICE.

Casual user keep the \$20 subscription

Heavy interactive hybrid: local loops, cloud for hard

Always-on agents buy the hardware